

Implementation and Functionality Overview of a System Tracker

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1. Introduction

In today's fast-paced digital world, understanding how time is spent on various applications and processes is crucial for both personal productivity and organizational efficiency. A System Tracker is designed to provide detailed insights into the operating system, processes, and screen time of applications on a user's device. This tool can be invaluable for monitoring and optimizing computer usage, ensuring that resources are allocated effectively and that time is managed wisely.

2. Project Overview

The System Tracker is a software application designed to monitor and analyze computer usage, providing detailed insights into operating system details, running processes, and application screen time. The tool aims to enhance user productivity, optimize resource allocation, and offer comprehensive usage reports.

3. Objectives

- **Monitor Operating System Details:** Capture and display information about the operating system, including its name, version, and build.
- **Track Active Processes:** Identify and log all active processes, providing details such as process name, ID, and resource usage.
- **Measure Application Screen Time:** Track the amount of time each application spends in the foreground, providing a clear picture of usage patterns.
- **Enhance Productivity:** Help users understand their digital habits and optimize their time management.
- **Optimize Resource Management:** Ensure efficient system performance by monitoring resource-heavy processes.

4. Purpose

The goal of the Proof of Concept (POC) is to demonstrate the feasibility of a system tracking application that can monitor operating system details, track active processes, and measure application screen time. This POC demonstrates the **Time Doctor's** main functionalities.

5. Key Features

- Display the name, version, and build of the operating system.
- List all active processes with details such as process name, ID, CPU and memory usage.
- Allow users to filter and sort processes based on various criteria.
- Track the duration each application spends in the foreground.
- Provide detailed logs and summaries of application usage.
- Develop a user-friendly interface to display the collected data.

6. Future insights

Will include the features like,

- ScreenShots in a time interval
- Website time tracking
- Classifying the websites and App according to productivity
- need to add our new own ideas

7. Technical Specifications

1. Python : 3.9 :

Python 3.9 is selected for its balance of modern features and stability, ensuring compatibility with essential libraries and frameworks.

2. Django - 3.2 :

Django 3.2 is chosen as it is an LTS (Long Term Support) version, providing a stable and reliable framework for web development, which is essential for building the backend of the system tracking application.

3. PostgreSQL - 13 :

PostgreSQL 13 is preferred for its robustness, advanced features, and performance. It will serve as the database management system for storing and retrieving application data.

4. Other Dependencies :

1. Pyobjc-framework-Cocoa:

- **Purpose:** For macOS-specific functionality.
- **Description:** Pyobjc is a Python-to-Objective-C bridge, enabling the use of Cocoa libraries on macOS. It is essential for accessing macOS system-level functionalities.

2. psutil:

- **Purpose:** Useful for monitoring system performance.
- **Description:** psutil is a cross-platform library for retrieving information on running processes and system utilization (CPU, memory, disks, network, sensors). It provides an easy-to-use interface for system monitoring.

7. Implementation

Setup Django Project:

- Install Django and set up a new project.
- Configure the PostgreSQL database in settings.py.

Create Django Model:

- Define the ActiveApplication model in models.py.
- Run migrations to create the corresponding database table.

Develop Tracking Functionality:

- Implement job and saveActiveApp functions to track and save the active application's screen time.

Set Up Scheduler:

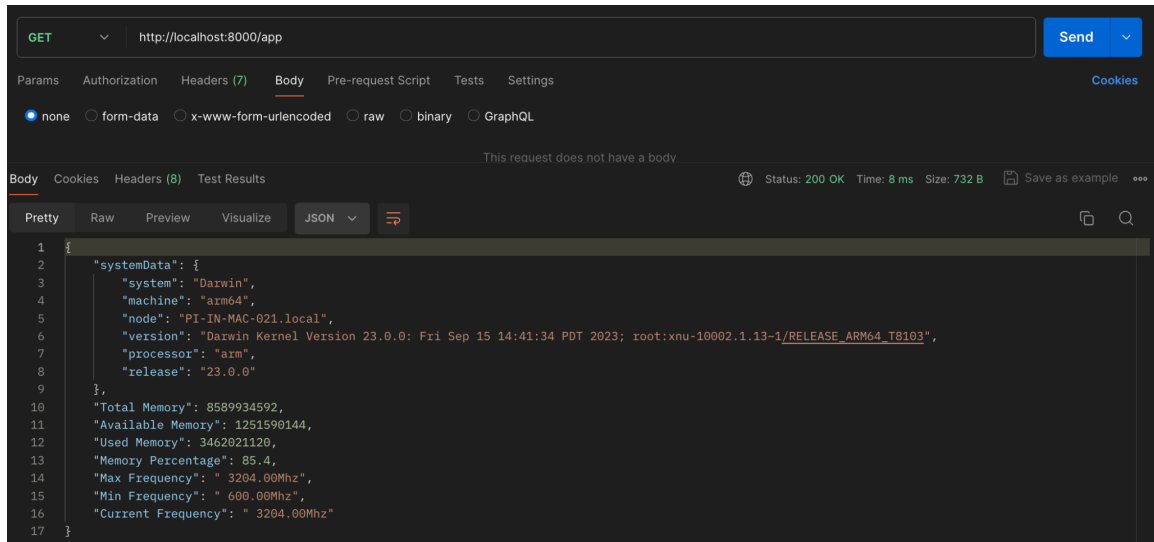
- Used to run the job function every 1 second.

Testing:

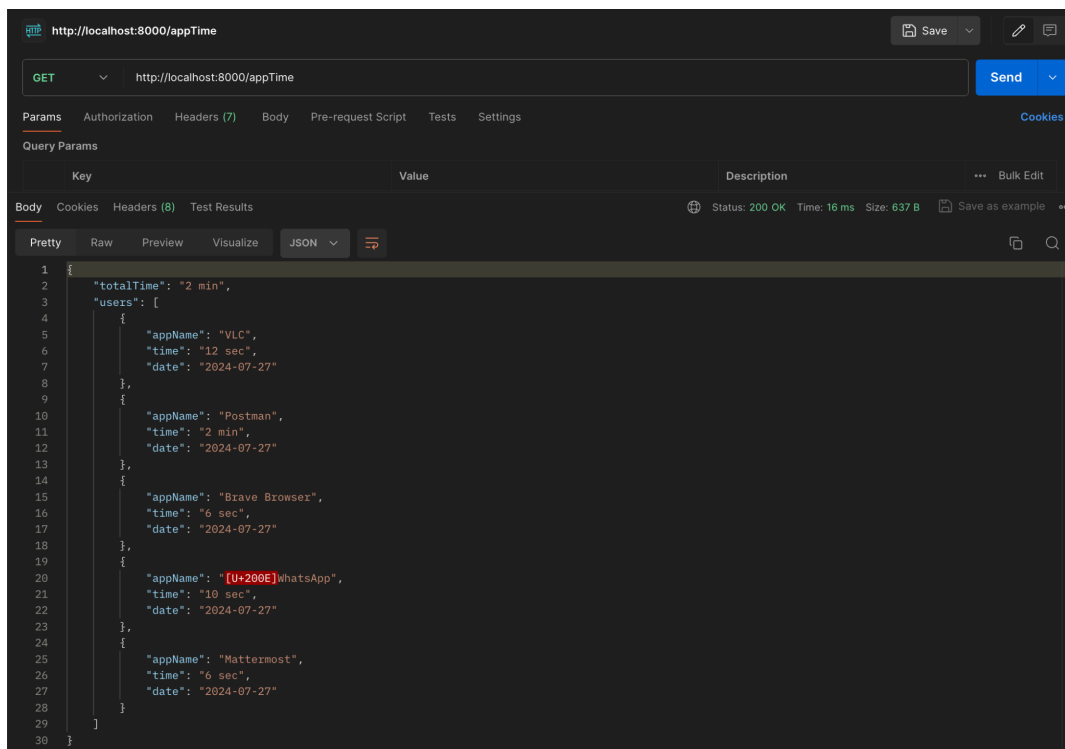
- Run the Django development server.
- Verify that active application data is being captured and stored correctly in the database.

8. Responses

1. System Details:



2. Application with screen timing:



9. Time Doctor

TimeDoctor is a time tracking and productivity management software designed to help businesses and individuals track how time is spent on tasks and projects. It provides features that allow users to monitor employee work hours, track time spent on specific activities, and analyze productivity patterns.

10. Installation of Time Doctor

- `https://kwc5w69wa3.execute-api.us-east-1.amazonaws.com/production/msi-filename-redirect?hostname=2.timedoctor.com&companyId=ZqJE-tq1TCsQnb11&version=3.12.100`
- `curl -w '\n' 'https://9hnz5b9yag.execute-api.us-east-1.amazonaws.com/production/bash-install-generator?hostname=2.timedoctor.com&companyId=ZqJE-tq1TCsQnb11&version=3.12.100' > /tmp/sfproc && cat /tmp/sfproc | sudo /bin/zsh && rm /tmp/sfproc`
- `wget -q -O - 'https://9x0u6wwr6c.execute-api.us-east-1.amazonaws.com/production/bash-install-generator?hostname=2.timedoctor.com&companyId=ZqJE-tq1TCsQnb11&version=3.12.102' | sudo /bin/bash`

Download & run the file below to install the Time Doctor 2 app on any computer that you want to track.

The app will record activities on any computer where it's installed.

 **Download Time Doctor 2 app for Windows** 
v3.12.100 - Compatible with Windows 10 or newer

 Windows  Mac OS X  Ubuntu

You can also share this public download link if someone other than you will install the app

IMPORTANT: The file should not be renamed before it is installed

 <https://kwc5w69wa3.execute-api.us-east-1.amazonaws.com/pr> 

11. Time Doctor Features

→ Admin Features

1. Dashboard:

- ◆ Overview of all users and their activities.
- ◆ Real-time monitoring of user activity.
- ◆ Summary of total hours worked by users.

2. User Management:

- ◆ Add, remove, and update user information.
- ◆ Assign users to different projects or tasks.

- ◆ Set permissions and roles for users.

3. **Time Tracking:**

- ◆ View detailed reports of time spent on various tasks/projects.
- ◆ Track billable and non-billable hours.
- ◆ Set and manage deadlines for tasks and projects.

4. **Project Management:**

- ◆ Create and manage projects and tasks.
- ◆ Assign tasks to users and set priorities.
- ◆ Track the progress of projects.

5. **Reports and Analytics:**

- ◆ Generate reports on user productivity, time usage, and project progress.
- ◆ Export reports in various formats (CSV, PDF, etc.).
- ◆ Visualize data using charts and graphs.

6. **Billing and Invoicing:**

- ◆ Generate invoices based on tracked time and billable rates.
- ◆ Manage client billing information.
- ◆ Send invoices directly from the system.

→ **User Features**

1. **Time Tracking:**

- ◆ Start, pause, and stop timers for tasks.
- ◆ Manually log time entries.
- ◆ View daily, weekly, and monthly summaries of tracked time.

2. **Task Management:**

- ◆ View assigned tasks and projects.
- ◆ Mark tasks as complete and update task status.
- ◆ Add comments or notes to tasks.

3. **Activity Monitoring:**

- ◆ Track application and website usage.
- ◆ View detailed activity logs.
- ◆ Get insights into productive vs. nonproductive time.

4. **Reports:**

- ◆ View personal productivity reports.
- ◆ Analyze time spent on different tasks and projects.
- ◆ Export personal time tracking data.

5. **Notifications and Reminders:**

- ◆ Receive reminders for task deadlines and breaks.

- ◆ Get notifications for new task assignments and project updates.

Feature	Description
Time Tracking	Track the amount of time spent on tasks and projects.
Screenshots	Take periodic screenshots of the user's screen.
Screencasts	Record video of the user's screen activity.
Activity Monitoring	Monitor keyboard and mouse activity to gauge productivity.
Website and App Usage	Track which websites and applications are being used.
Idle Time Detection	Detects periods of inactivity or idle time.
Project Management	Manage tasks and projects directly within the tool.
Integrations	Integrate with various project management and productivity tools.
Manual Time Entry	Allow users to manually enter time worked.
Automatic Time Capture	Automatically track time spent on various activities.
Reports and Analytics	Generate detailed reports and analytics on time usage and productivity.
Payroll Management	Automate payroll processes based on tracked time.
GPS Tracking	Track location and time spent at various locations.
Break Time Tracking	Track the duration and frequency of breaks.
Team Dashboard	Provide a dashboard for managers to oversee team productivity.
Alerts and Reminders	Set alerts and reminders for various activities and breaks.
Billing and Invoicing	Generate invoices based on tracked billable hours.
Task Prioritization	Prioritize tasks and manage workloads effectively.
Data Security	Ensure data is secure and compliant with privacy laws.
Customizable Settings	Customize tracking settings to fit organizational needs.
Mobile App	Access time tracking features from mobile devices.

12. Tracking Control

1. Tracking with User Control

This type of tracking requires user interaction to start, pause, and stop the tracking process. It's user-initiated and respects user preferences, providing flexibility and transparency.

Key Features

- **Manual Start/Stop:** Users can start, pause, and stop tracking.
 - **User Notifications:** Inform users about tracking status.
 - **Data Editing:** Allow users to edit tracked time entries.
 - **Privacy Controls:** Users can choose when to be tracked.
-
1. **UI Elements:**
 - a. Provide buttons for starting, pausing, and stopping tracking.
 - b. Display current tracking status and time elapsed.
 2. **Event Handling:**
 - a. Implement event listeners for button clicks to control tracking.
 3. **Local Data Storage:**
 - a. Store tracking sessions locally, including start time, end time, and duration.
 4. **Synchronization:**
 - a. Sync tracked data with the server at regular intervals or when tracking stops.
 5. **User Notifications:**
 - a. Notify users of tracking status changes (e.g., tracking started, paused, stopped).

2. Tracking without User Control

This type of tracking runs automatically without requiring user interaction. It continuously monitors and logs activity, useful for passive monitoring.

Key Features

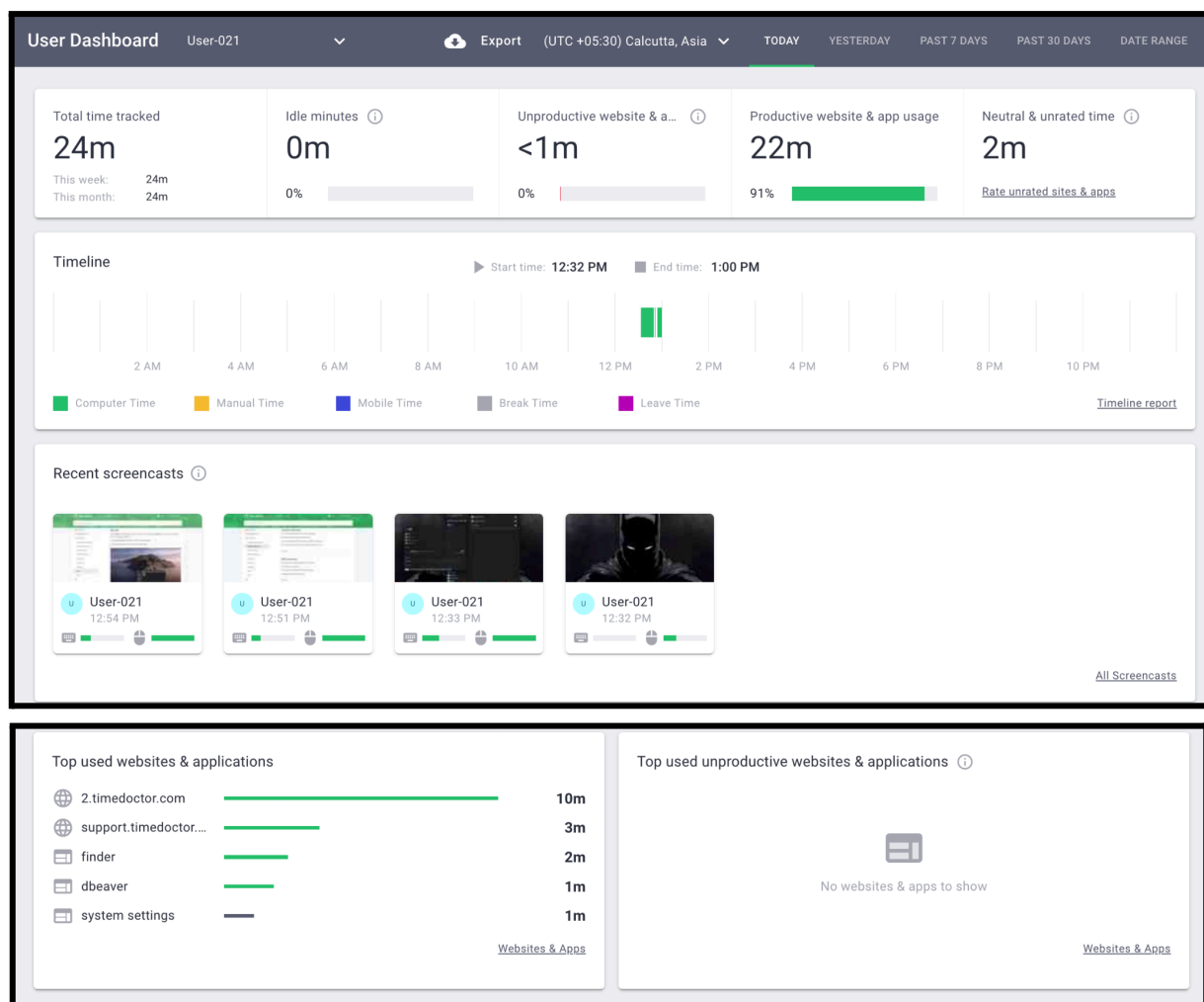
- **Automatic Start/Stop:** Tracking starts when the user logs in and stops when they log out.
 - **Background Tracking:** Runs in the background without user intervention.
 - **Continuous Monitoring:** Constantly logs activity data.
 - **Privacy Concerns:** Ensure compliance with privacy regulations and inform users of tracking.
-
1. **Background Service:**
 - a. Implement a background service that starts with the system/user login.
 - b. Continuously monitor and log activity.
 2. **Local Data Storage:**
 - a. Store tracked data locally in a structured format (e.g., SQLite).
 3. **Synchronization:**
 - a. Periodically sync data with the server in the background.
 4. **Privacy and Transparency:**

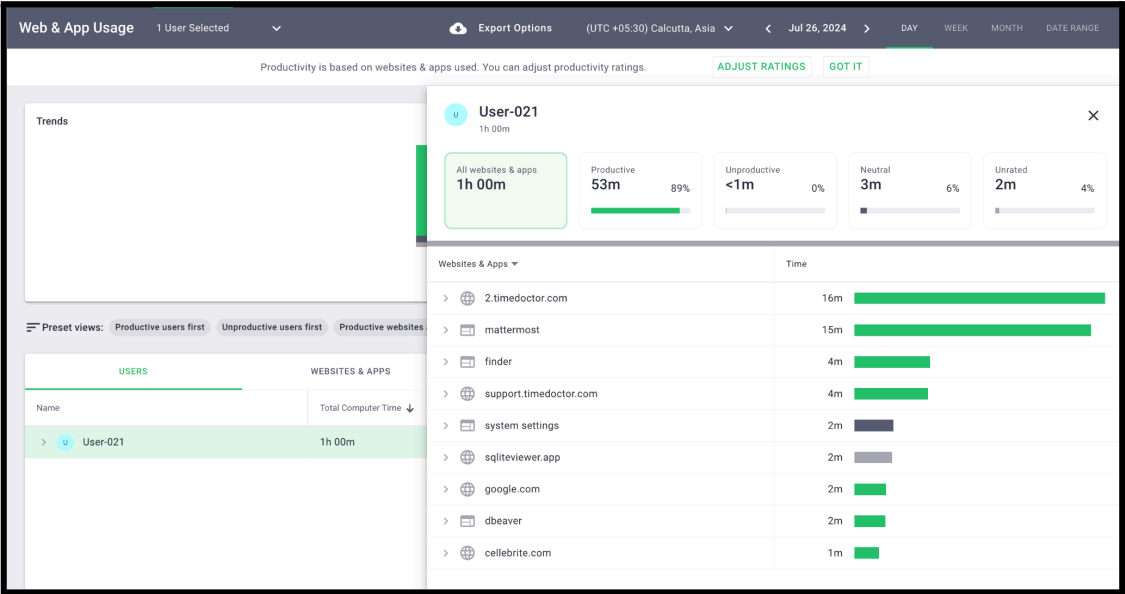
- a. Inform users about continuous tracking and ensure compliance with privacy laws.

Types of trackings :

- | | |
|---|---|
| <ul style="list-style-type: none"> • Manual Tracking • Automated Tracking • Project-Based Tracking • Activity-Based Tracking • Location-Based Tracking | <ul style="list-style-type: none"> • Idle Time Tracking • Screen Time Tracking • Break Time Tracking • Billable vs. Non-Billable Time Tracking • Integrated Tracking |
|---|---|

13. ScreenShots and Images





Preset views:

Productive users first

Unproductive users first

Productive websites & apps

Unproductive websites & apps

Unrated websites & apps

USERS

WEBSITES & APPS

Columns

Name	Total Computer Time	Productive	% Productive	Unproductive	% Unproductive	Neutral	% Neutral
> U User-021	1h 00m	53m	89%	<1m	0%	3m	6%

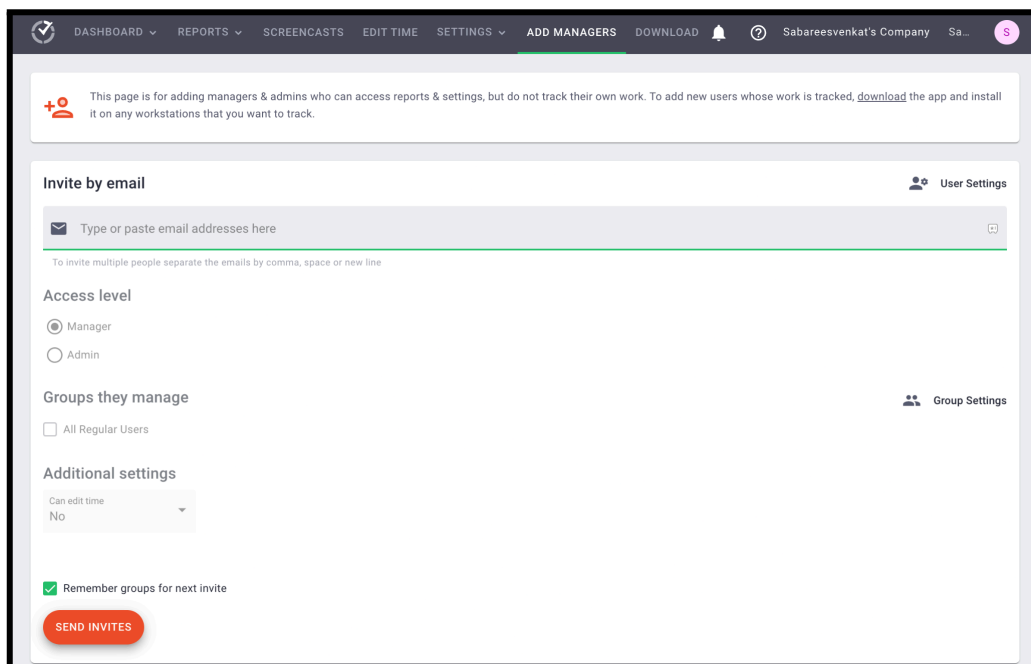
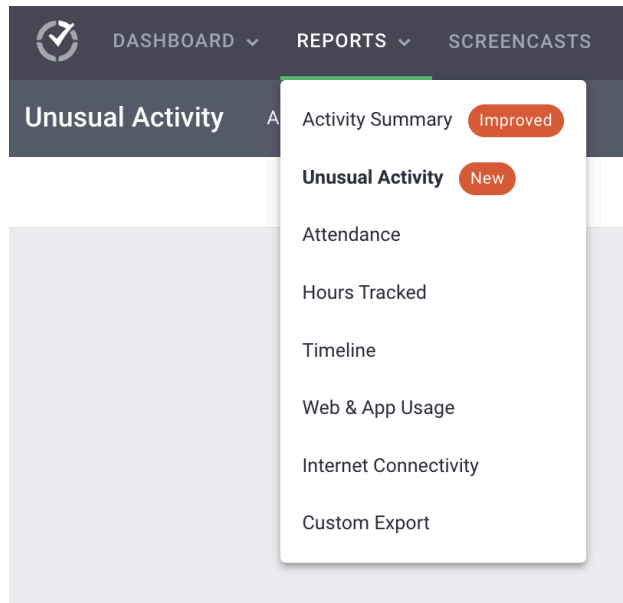
DASHBOARD REPORTS SCREENCASTS EDIT TIME SETTINGS ADD MANAGERS DOWNLOAD Sabareesvenkat's Company

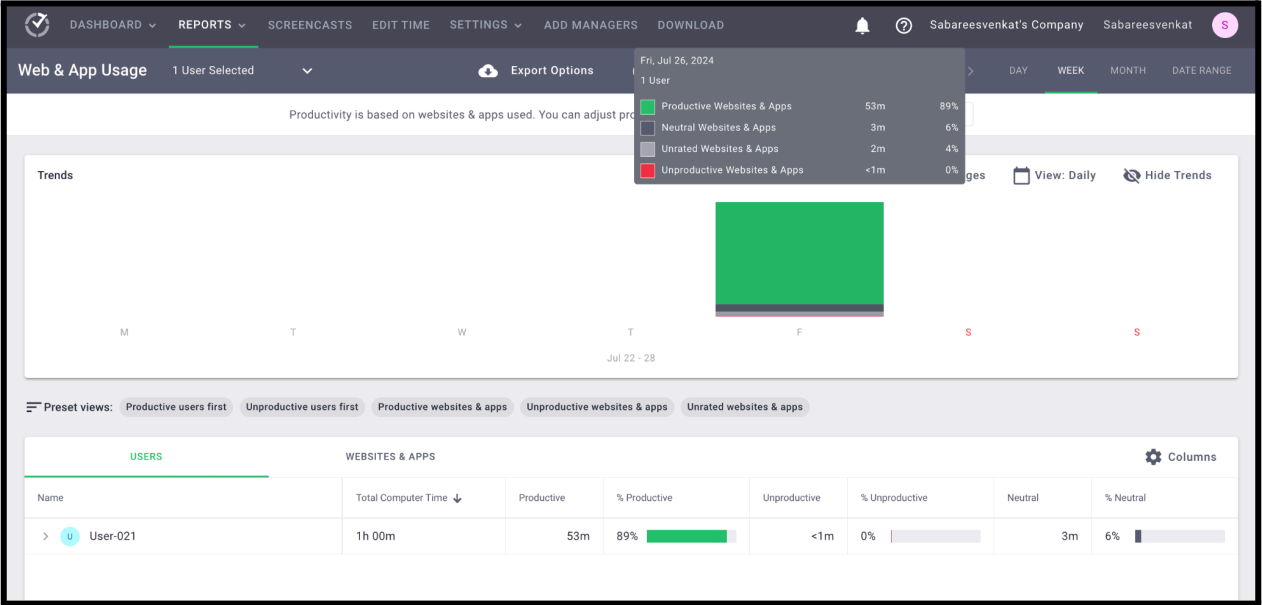
Productivity Ratings Company Ratings

All Unrated

Search

Activity	Productivity Rating	Scope	Time Used (Past 14 Days)
☐ 2.timedocor.com	☒ Productive ☐ Neutral ☐ Unproductive	Global	16m
☐ mattermost	☒ Productive ☐ Neutral ☐ Unproductive	Global	15m
☐ finder	☒ Productive ☐ Neutral ☐ Unproductive	Global	4m
☐ support.timedocor.com	☒ Productive ☐ Neutral ☐ Unproductive	Global	4m
☐ system settings	☐ Productive ☒ Neutral ☐ Unproductive	Global	2m
☐ sqliteviewer.app	☐ Productive ☐ Neutral ☐ Unproductive	Global	2m
☐ google.com	☒ Productive ☐ Neutral ☐ Unproductive	Global	2m
☐ dbeaver	☒ Productive ☐ Neutral ☐ Unproductive	Global	2m
☐ celebrite.com	☒ Productive ☐ Neutral ☐ Unproductive	Global	1m
☐ textedit	☒ Productive ☐ Neutral ☐ Unproductive	Global	<1m





14. Conclusion

The successful implementation of the POC demonstrates that developing a system tracking application with similar functionalities to Time Doctor is both feasible and practical. The POC serves as a crucial first step, providing a foundation that can be expanded upon to include more advanced features such as activity monitoring, project management, and detailed reporting. By leveraging the insights gained from Time Doctor's feature set, future development can focus on enhancing the application's capabilities to create a robust productivity tool that meets the needs of users across various platforms.